

Retrospective EHR Post-Discharge Utilization Analytics Report

Abstract

This report presents a retrospective healthcare analytics workflow using Synthea synthetic EHR data. The analysis defines an adult inpatient cohort, selects the first eligible inpatient encounter as the index hospitalization, derives post-discharge outpatient follow-up timing and ED revisit measures, identifies all-cause 30-day inpatient readmission, validates cohort and temporal logic, and produces aggregate reporting outputs for healthcare operations and research analytics audiences. The final synthetic cohort included 255 adult patients. The 30-day inpatient readmission rate was 5.1%, outpatient follow-up within 30 days was 21.6%, and ED revisit within 30 days was 0.8%. Prior utilization burden and prior ED use were the clearest descriptive signals in the cohort. Exploratory logistic regression was included as a secondary demonstration analysis and should not be interpreted as clinical evidence. The project uses synthetic data only and is not intended for clinical decision-making.

Executive Summary

This report summarizes a validated retrospective healthcare analytics workflow using Synthea synthetic EHR data. The analysis focuses on adult inpatient cohort definition, post-discharge utilization tracking, outpatient follow-up timing, ED revisits, and all-cause 30-day inpatient readmission.

The final analytic cohort included 255 adult patients with one index inpatient encounter per patient. The 30-day inpatient readmission rate was 5.1%. Outpatient follow-up within 30 days occurred for 21.6% of patients and is interpreted as an observed utilization measure, not as evidence that follow-up prevents readmission.

The clearest descriptive signals were prior utilization burden and prior ED use. Patients with readmission had higher prior encounter counts and higher prior ED visit counts than patients without readmission. Stratified sensitivity outputs also showed higher observed readmission rates among patients with high prior utilization or any prior ED use. Exploratory logistic regression was included to demonstrate adjusted association reporting, but the small event count limits interpretation and does not support clinical prediction.

Research Question

Among adult patients with a first eligible acute inpatient hospitalization in Synthea synthetic EHR data, how are outpatient follow-up timing, demographic characteristics, clinical conditions, prior utilization, and discharge-related factors associated with all-cause inpatient readmission within 30 days of discharge?

Data Source

The project uses Synthea synthetic EHR CSV files stored locally in `data/raw/`. The report uses aggregate validation tables, analysis outputs, and static report figures generated from the synthetic workflow. No real patient data are included.

Methods Summary

The SQL and Python workflow profiles source encounters, defines eligible adult inpatient encounters, selects the first eligible inpatient encounter as the index hospitalization, derives post-discharge utilization measures, identifies all-cause 30-day inpatient readmission, validates cohort and temporal logic, and generates report-ready aggregate outputs. The workflow separates cohort construction, validation, descriptive analysis, sensitivity analysis, and exploratory modeling so each analytic step can be reviewed independently.

Workflow Architecture

The project follows a validation-first SQL and Python workflow. Raw Synthea CSV files remain local, while the report uses aggregate validation outputs, analysis tables, and static figures to communicate cohort construction, post-discharge utilization, and readmission findings.

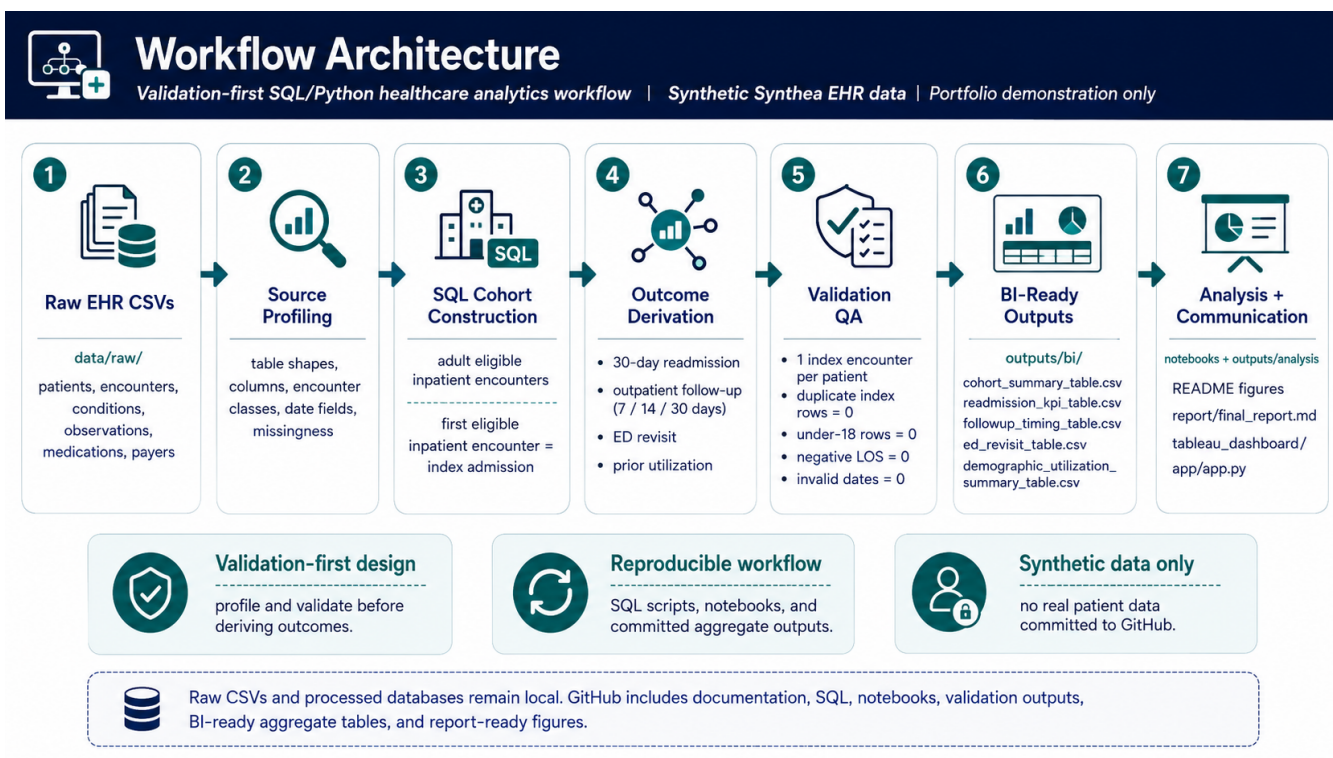


Figure 1. Workflow architecture

Synthetic Synthea EHR cohort | Static portfolio summary visual | Not for clinical decision-making



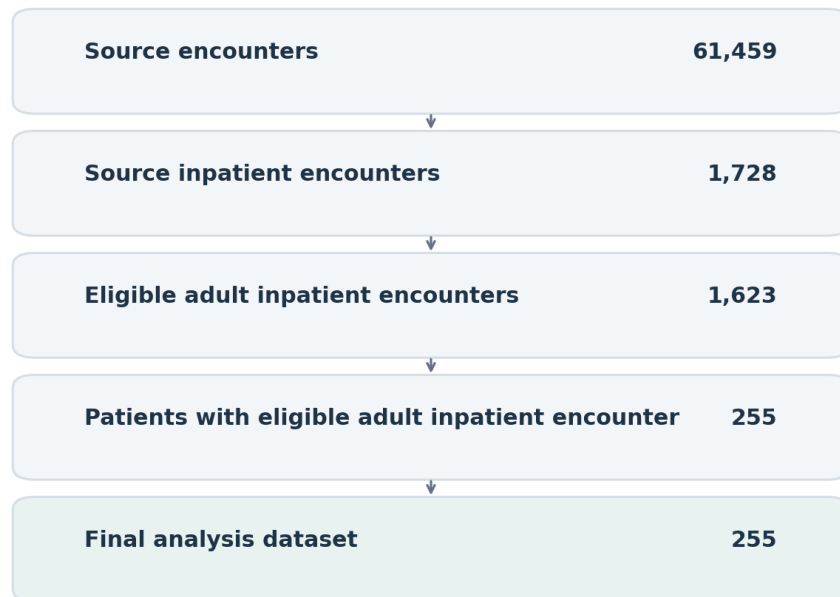
Figure 2 summarizes the core analytic story in one view: cohort attrition, timing of readmission, post-discharge utilization, outpatient follow-up accumulation, and prior utilization differences by readmission status. The most important takeaway is that prior encounter burden and prior ED use stand out more clearly than outpatient follow-up as descriptive signals in this synthetic cohort.

Cohort Construction

The final analysis dataset includes one index encounter per adult patient with a first eligible inpatient encounter. Cohort construction began with 61,459 source encounters and ended with 255 adult patients in the final analytic dataset.

Cohort step	Count
Source patients	1,163
Source encounters	61,459
Source inpatient encounters	1,728
Eligible adult inpatient encounters	1,623
Patients with eligible adult inpatient encounter	255
Index encounters	255
Final analysis dataset rows	255

Figure 1. Cohort attrition flow diagram



Counts are derived from aggregate validation outputs. Synthetic data only.

Figure 3. Cohort attrition flow diagram

Encounter Classification

The source encounters.csv file contains distinguishable encounter classes needed for the inpatient cohort and post-discharge utilization definitions.

Encounter class	Count	Percent
wellness	24,038	39.11
ambulatory	20,124	32.74
outpatient	10,837	17.63

Encounter class	Count	Percent
urgentcare	2,564	4.17
emergency	2,168	3.53
inpatient	1,728	2.81

Validation Findings

Validation checks support the current SQL cohort logic:

- One index encounter per patient: 255 index rows and 255 distinct patients.
- Duplicate patient index rows: 0.
- Missing patient IDs in final dataset: 0.
- Missing index encounter IDs in final dataset: 0.
- Under-18 index rows: 0.
- Negative length-of-stay rows: 0.
- Discharge-before-admission rows: 0.
- Missing or invalid encounter start/stop dates in final dataset: 0.

Missingness is expected for timing variables that only apply to patients with observed readmission or outpatient follow-up. For example, `days_to_readmission` is missing for patients without a 30-day readmission, and `days_to_first_outpatient_followup` is missing for patients without observed outpatient follow-up.

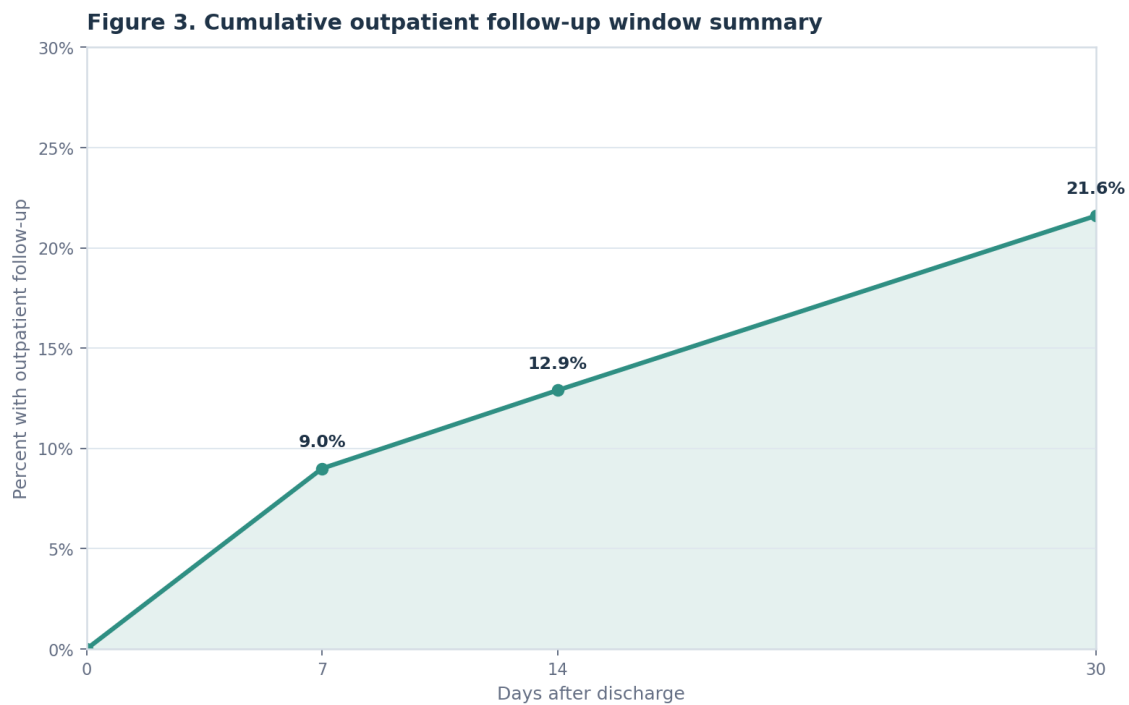
Static Reporting Outputs

The project includes report-ready aggregate tables and static summary visuals that support stakeholder-facing interpretation of readmission, follow-up timing, ED revisits, and cohort characteristics.

Aggregate Results Snapshot

Measure	Value
Final analysis cohort	255 patients
30-day inpatient readmission	13 patients
30-day inpatient readmission rate	5.1%
Mean days to readmission	20.98
Outpatient follow-up within 7 days	23 patients, 9.0%
Outpatient follow-up within 14 days	33 patients, 12.9%
Outpatient follow-up within 30 days	55 patients, 21.6%
ED revisit within 30 days	2 patients, 0.8%
Any post-discharge encounter within 30 days	96 patients, 37.6%

Figure 4 shows that outpatient follow-up accumulates gradually across the 30-day window. These follow-up measures describe observed post-discharge utilization and should not be interpreted causally.



Cumulative window summary based on aggregate follow-up outputs at 7, 14, and 30 days.

Figure 4. Cumulative outpatient follow-up curve

Descriptive Analysis Outputs

The analysis outputs include baseline characteristics, readmission summaries, outpatient follow-up summaries, ED revisit summaries, risk stratification tables, prior utilization stratification, prior ED use stratification, follow-up window sensitivity, and model sensitivity results. Table 1 compares baseline characteristics by 30-day readmission status using aggregate summaries only.

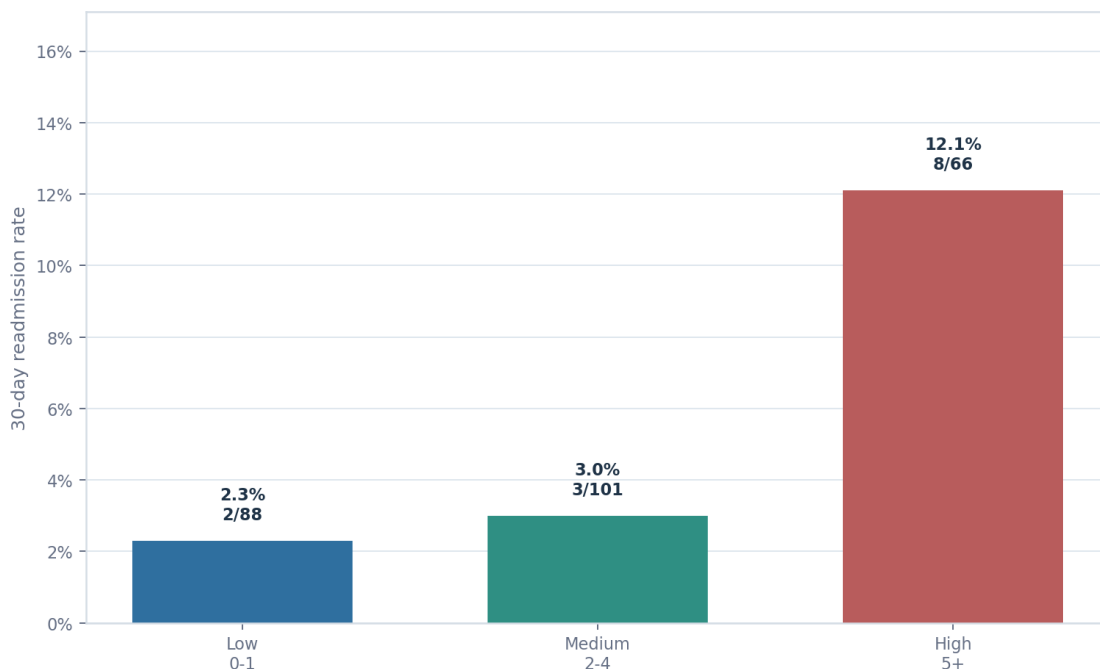
Analytic Findings

Prior utilization was the clearest descriptive signal in the synthetic cohort:

- Prior encounters in the 12 months before index admission were higher among readmitted patients: 4.92 versus 3.66 encounters, $p = 0.0405$.
- Prior ED visits in the 12 months before index admission were higher among readmitted patients: 0.69 versus 0.24 visits, $p = 0.0005$.
- Outpatient follow-up within 30 days was similar between groups: 23.1% among readmitted patients versus 21.5% among non-readmitted patients.
- ED revisit within 30 days was rare overall and should not be overinterpreted.

These findings suggest that prior ED use and broader prior utilization may be useful operational signals for transition-of-care review, outreach prioritization, or further subgroup analysis. Age-group and condition-group patterns remain exploratory because several strata have small event counts.

Figure 4. Readmission rate by prior utilization group



Descriptive synthetic-data output. Prior utilization groups are based on prior encounter counts in the 12 months before index admission.

Figure 5. Readmission rate by prior utilization group

Statistical Caution for Subgroup Outputs

The subgroup outputs include uncertainty fields for age-group and condition-group readmission summaries. These fields are intended to make subgroup figures more transparent, not to convert the synthetic-data analysis into clinically validated inference.

For age group, 30-day readmission was 5.8% among patients under 65, with a Wilson 95% confidence interval of 3.3% to 9.8%, and 2.1% among patients age 65 or older, with a Wilson 95% confidence interval of 0.4% to 11.1%. Fisher's exact test was used for the age-group comparison because at least one expected cell count was less than 5; the exploratory comparison p-value was 0.4726.

For condition groups, readmission rates were exported with Wilson 95% confidence intervals and subgroup sample-size notes:

Condition group	Patients	Readmitted	Readmission rate	Wilson 95% CI	Sample-size note
COPD	4	0	0.0%	0.0%-49.0%	Small subgroup; rate should be interpreted cautiously.
Chronic kidney disease	8	1	12.5%	2.2%-47.1%	Small subgroup; rate should be interpreted cautiously.
Diabetes	21	2	9.5%	2.7%-28.9%	Small subgroup; rate should be interpreted cautiously.
Hypertension	74	4	5.4%	2.1%-13.1%	Subgroup size ≥ 30 ; still descriptive and exploratory.

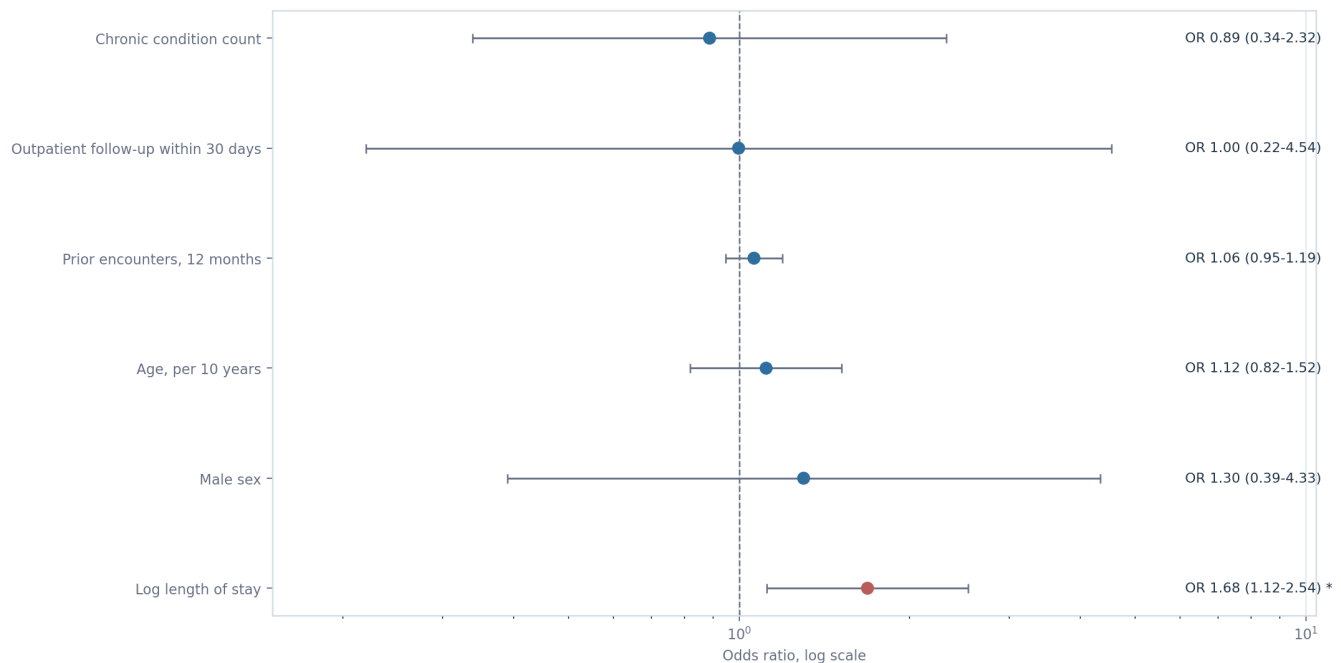
These subgroup results are based on synthetic Synthea data and simplified condition flags. They should be interpreted as exploratory reporting context only. They are not causal findings, clinically representative estimates, or a validated risk stratification model.

Exploratory Logistic Regression

An exploratory logistic regression model was fit for 30-day inpatient readmission using age, sex, log length of stay, prior encounters, chronic condition count, and outpatient follow-up within 30 days. The model included 255 observations and 13 readmission events, so estimates may be unstable and should be interpreted cautiously. Log length of stay was the only statistically significant predictor, but the model should not be interpreted as clinically valid or predictive.

Predictor	Odds ratio	95% CI	p-value
Age, per 10 years	1.1158	0.8201-1.5180	0.4855
Male sex	1.2994	0.3896-4.3341	0.6700
Log length of stay	1.6847	1.1189-2.5365	0.0125
Prior encounters, 12 months	1.0621	0.9457-1.1929	0.3092
Chronic condition count	0.8868	0.3387-2.3218	0.8067
Outpatient follow-up within 30 days	0.9978	0.2193-4.5402	0.9977

Figure 5. Exploratory Logistic Regression for 30-Day Readmission



Synthetic-data demonstration only; estimates are exploratory and limited by small event count. Red marker indicates $p < 0.05$.

Figure 6. Exploratory logistic regression forest plot

These model results are secondary to the descriptive and stratified findings. In this synthetic cohort, the main analytic story remains prior utilization burden and prior ED use, not model-based prediction.

Sensitivity and Stratified Analysis

Sensitivity and stratified analyses were used to test whether the descriptive story changed across practical utilization groups and follow-up timing windows. These analyses are intended for healthcare operations interpretation, not causal inference.

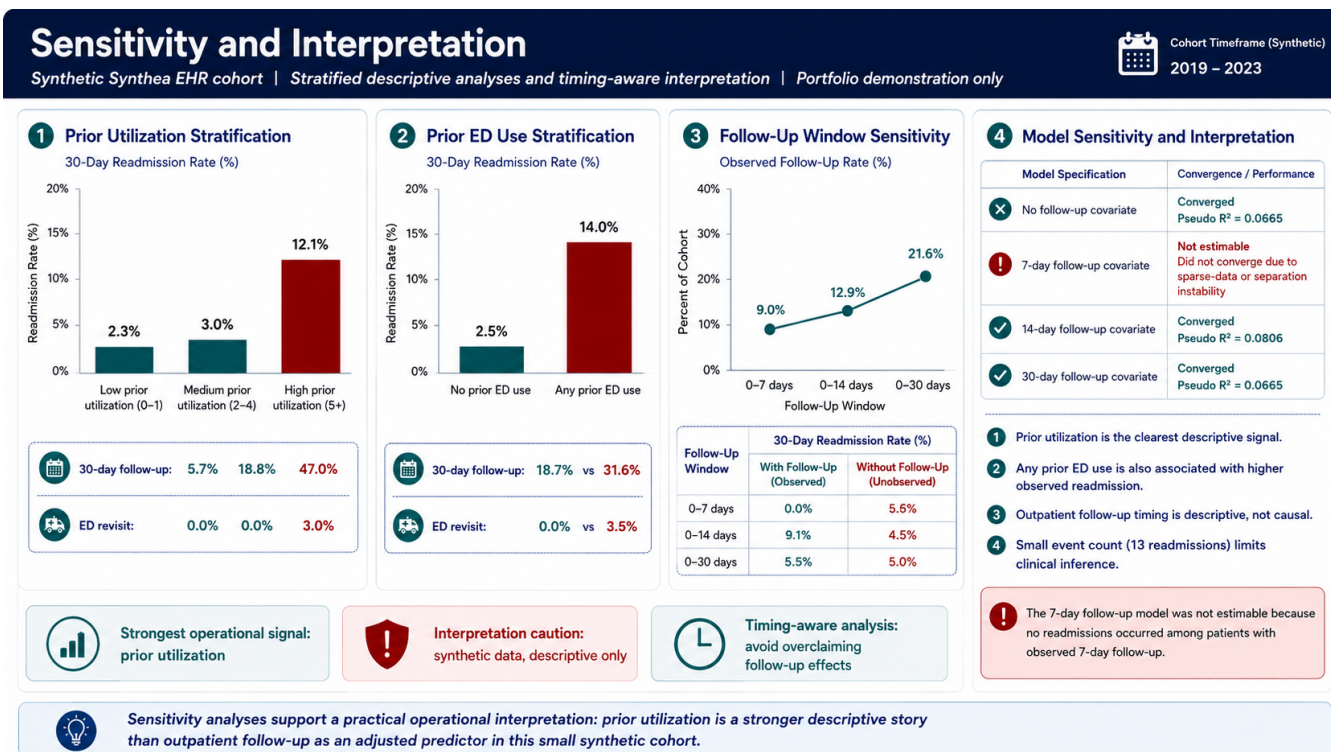


Figure 7. Sensitivity and interpretation summary visual

Prior Utilization Stratification

Prior utilization group	Patients	Readmitted	Readmission rate	30-day follow-up	ED revisit
Low prior utilization, 0-1 encounters	88	2	2.3%	5.7%	0.0%
Medium prior utilization, 2-4 encounters	101	3	3.0%	18.8%	0.0%
High prior utilization, 5+ encounters	66	8	12.1%	47.0%	3.0%

Patients with higher prior encounter burden had higher observed readmission, follow-up, and ED revisit rates. In a real health system setting, this type of table could support discussion about whether prior utilization should inform transition-of-care review or outreach prioritization.

Prior ED Use Stratification

Prior ED use group	Patients	Readmitted	Readmission rate	30-day follow-up	ED revisit
No prior ED use	198	5	2.5%	18.7%	0.0%
Any prior ED use	57	8	14.0%	31.6%	3.5%

Patients with any prior ED use had a higher observed readmission rate than patients without prior ED use. This should be interpreted as a descriptive utilization signal from synthetic data, not a validated risk factor.

Follow-Up Window Sensitivity

Follow-up window	Follow-up count	Follow-up percent	Readmission rate with follow-up	Readmission rate without follow-up
0-7 days	23	9.0%	0.0%	5.6%
0-14 days	33	12.9%	9.1%	4.5%
0-30 days	55	21.6%	5.5%	5.0%

Observed outpatient follow-up increased as the window widened. These results describe observed follow-up patterns only. The follow-up groups are not randomized, and timing may be affected by early readmission, illness severity, care access, and scheduling processes.

Model Sensitivity

Model	Follow-up variable	Converged	Pseudo R-squared	AIC	LLR p-value
No follow-up covariate	None	Yes	0.0665	107.88	0.2336
With 7-day follow-up covariate	7-day follow-up	No	Not estimable	Not estimable	Not estimable
With 14-day follow-up covariate	14-day follow-up	Yes	0.0806	108.43	0.2181
With 30-day follow-up covariate	30-day follow-up	Yes	0.0665	109.88	0.3368

The 7-day follow-up model could not be estimated because no readmissions occurred among patients with observed 7-day outpatient follow-up, producing sparse-data or separation instability. Overall, the sensitivity analysis supports a practical interpretation: prior utilization is a stronger descriptive signal than outpatient follow-up as an adjusted predictor in this small synthetic cohort.

Final Interpretation

The final analytic story is that prior healthcare utilization is the most useful operational signal in this synthetic post-discharge cohort. Patients with high prior encounter burden and patients with any prior ED use had higher observed 30-day readmission rates than lower-utilization groups. This supports a realistic healthcare analytics interpretation: prior utilization could be used as a stakeholder-facing stratification variable for transition-of-care review, care management discussion, or further cohort refinement.

Outpatient follow-up timing remains important to measure, but it should be interpreted cautiously. Follow-up within 7, 14, and 30 days describes observed post-discharge utilization patterns; it does not show whether follow-up prevents readmission. The sensitivity models reinforce that caution, because the 7-day follow-up model was not estimable in this small synthetic cohort due to sparse-data or separation instability.

The project demonstrates a realistic retrospective healthcare analytics workflow: validated cohort construction, post-discharge outcome derivation, utilization stratification, sensitivity analysis, and stakeholder-facing interpretation without overclaiming clinical validity.

Translation to a Real Health System Setting

In a real hospital or health system analytics environment, this workflow would map to a common post-discharge utilization request from a physician investigator, quality improvement leader, or population health team. The same structure could be adapted to governed EHR warehouse tables after data access approval, privacy review, clinical validation of cohort definitions, and comparison against existing operational reporting logic.

The strongest practical use would be identifying whether prior utilization and prior ED use should support transition-of-care review or outreach prioritization. The analysis would also help stakeholders distinguish descriptive utilization monitoring from causal claims about whether outpatient follow-up prevents readmission.

Limitations

- Synthea records are synthetic and are not real patient records.
- The encounter and condition distributions may not reflect a real health system population.
- The current report is based on aggregate outputs and does not include patient-level review.
- Condition flags use simplified grouping logic suitable for synthetic EHR data.
- Planned versus unplanned readmission distinctions are not implemented in the current MVP.
- The logistic regression model is exploratory and limited by the small number of synthetic readmission events.
- The project is not intended for clinical decision-making, quality reporting, or operational deployment.

Reproducibility

Detailed reproduction steps are better suited for the project README or technical documentation. The main report focuses on the analytic question, validated cohort logic, aggregate findings, sensitivity analyses, and interpretation.